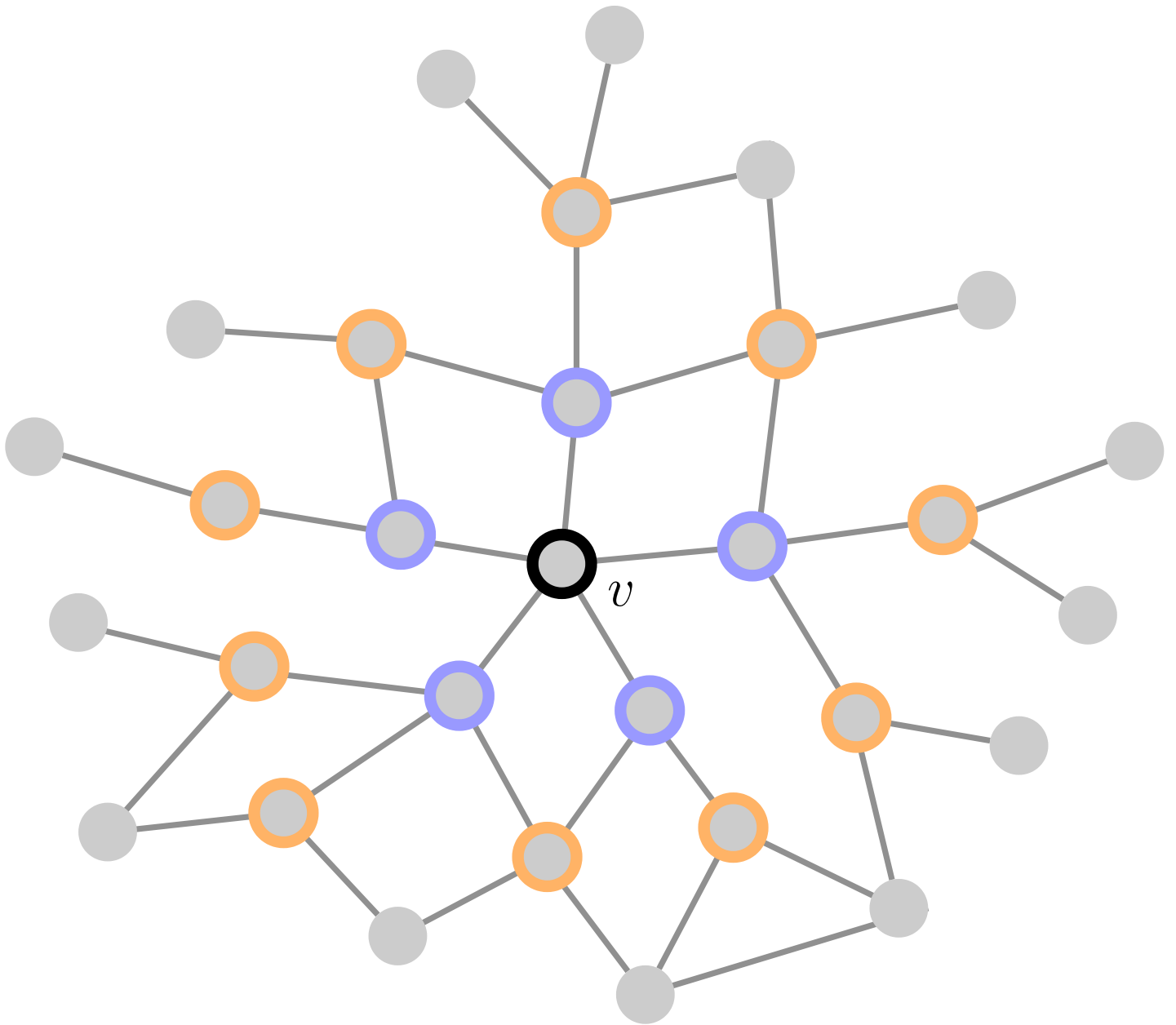




$$g(v) = [\varphi(\mathcal{X}_v^1), \varphi(\mathcal{X}_v^2)] \in \mathbb{R}^8$$

$$|\vec{x}_v\rangle \approx \begin{bmatrix} g^T(v) \\ \dots \end{bmatrix}$$

$$\frac{1}{N} \sum_{v=0}^{N-1} |v\rangle |0\rangle \rightarrow \frac{1}{N} \sum_{v=0}^{N-1} |v\rangle |\vec{x}_v\rangle$$